$$\begin{bmatrix} 0.2 & 0.1 & 0.7 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0.3 & 0 & 0 & 0.3 & 0.4 \\ 0 & 0.5 & 0 & 0.3 & 0.2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

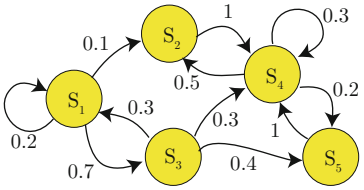


Fig. 6.6 A Markov matrix (*left*) and its graph representation (*right*). In this example, the Markov Chain is not irreducible

some integers l_1 and l_2. This means that it is possible to reach j from i with positive probability and vice-versa.

A finite state space model is *irreducible* if every state is reachable from every other state in a finite "time", where time in this context refers to the number of iterations of the chain. Equivalently, we say the model is irreducible if all states communicate.

A very useful representation of the Markov Chain transition matrix $\boldsymbol{\Pi}$ is by use of a weighted directed graph in which each node represents a state of the Markov Chain and the weights represent transition probabilities. Then one may understand irreducibility as expressing the connectedness[8] of this graph (see Fig. 6.6).

An *invariant distribution* is a left eigenvector of $\boldsymbol{\Pi}$, having only non-negative elements, and corresponding to the eigenvalue 1. That is, an invariant distribution $\boldsymbol{\psi}_*$ satisfies

$$\boldsymbol{\psi}_* = \boldsymbol{\psi}_* \boldsymbol{\Pi}\,.$$

The fundamental result for Markov chains is the following: *If a given Markov chain is both irreducible and aperiodic, then it can be shown that there is a unique invariant distribution* $\boldsymbol{\psi}_*$ *such that*

$$\lim_{n\to\infty} \boldsymbol{\psi}_n = \boldsymbol{\psi}_*.$$

for arbitrary initial distribution $\boldsymbol{\psi}_0$. The convergence of an irreducible finite Markov chain to its stationary distribution can be understood through the properties of its eigenvalues by applying of the Perron-Frobenius theorem (see [269]).

[8] We say that a graph is *connected* if every node is linked to every other node by a sequence of contiguous edges. For a directed graph, we must have, for any pair of vertices A and B, paths which take us from A to B and vice-versa.